THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS UNIT TEST SET B

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | RBSE Board Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures; use graph paper where a graph is asked.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. The sum of the first 10 terms of $2, 4, 6, \ldots$ is: 1

(a) 110 **(b)** 90 **(c)** 100 **(d)** 120

2. The number of terms in the AP $7, 13, 19, \ldots, 205$ is: 1

(a) 32 **(b)** 33 **(c)** 35 **(d)** 34

3. If $k+2,\ 4k-6,\ 3k-2$ are consecutive terms of an AP, then k is: 1

(a) 4 **(b)** 5 **(c)** 3 **(d)** 2

4. A student states: "The sum of the first n odd natural numbers is $n(n+1)$." The correct response is: 1

(a) Correct **(b)** Correct only for even n

(c) Incorrect; the sum is n^2 **(d)** Cannot be determined

5. In $\triangle ABC$, $DE \parallel BC$ with $AD = 2$ cm, $DB = 3$ cm and $AE = 4$ cm. Then EC is: 1

(a) 4 cm **(b)** 8 cm **(c)** 5 cm **(d)** 6 cm

6. A vertical stick 12 m long casts a shadow 8 m long. At the same time a tower casts a shadow 40 m long. The height of the tower is: 1

(a) 60 m **(b)** 45 m **(c)** 80 m **(d)** 50 m

7. If $\triangle PQR \sim \triangle XYZ$ with $\angle Q = 50^\circ$ and $\angle R = 70^\circ$, then $\angle X$ is: 1

(a) 60° **(b)** 50° **(c)** 70° **(d)** 120°

8. $\triangle ABC \sim \triangle DEF$ with $\frac{AB}{DE} = \frac{2}{3}$. If the perimeter of $\triangle ABC$ is 24 cm, the perimeter of $\triangle DEF$ is: 1

(a) 54 cm **(b)** 36 cm **(c)** 48 cm **(d)** 16 cm

9. The fourth vertex D of parallelogram ABCD with A(1, 2), B(3, 5), C(7, 7) is: 1

(a) $(9, 10)$ **(b)** $(5, 4)$ **(c)** $(3, 4)$ **(d)** $(4, 5)$

10. The distance of the point $(5, -12)$ from the origin is: 1

(a) 12 **(b)** 17 **(c)** 7 **(d)** 13

11. The point dividing the join of $(1, 3)$ and $(4, 6)$ internally in the ratio 2 : 1 is: 1

(a) $(3, 4)$ **(b)** $(3, 5)$ **(c)** $(2, 4)$ **(d)** $(2, 5)$

12. If the mid-point of the segment joining $(x, 4)$ and $(6, y)$ is $(4, 5)$, then (x, y) is: 1

(a) $(2, 6)$ **(b)** $(4, 6)$ **(c)** $(6, 2)$ **(d)** $(2, 5)$

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

13. Find the 31st term of an AP whose 11th term is 38 and 16th term is 73. 2

14. Find the sum of the odd numbers between 0 and 50. 2

15. State the AA similarity criterion. Are two right triangles, each having an acute angle of 35°, similar? Why? 2

16. Find the point on the y-axis which is equidistant from A(6, 5) and B(−4, 3). 2

17. Find the value of x if the distance between $(x, -1)$ and $(3, 2)$ is 5. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

18. Find the sum: $34 + 32 + 30 + \cdots + 10$. 3

19. ABCD is a trapezium in which $AB \parallel DC$ and its diagonals intersect each other at O. Show that $\frac{AO}{BO} = \frac{CO}{DO}$. 3

20. Find the ratio in which the y-axis divides the segment joining $(5, -6)$ and $(-1, -4)$. Also find the point of intersection. 3

21. Find the type of quadrilateral formed by A(−1, −2), B(1, 0), C(−1, 2) and D(−3, 0). Give reasons. 3

22. D is a point on side BC of $\triangle ABC$ such that $\angle ADC = \angle BAC$. Show that $CA^2 = CB \cdot CD$. 3

OR

A 1.5 m tall boy stands 2.4 m away from a lamp-post and his shadow is 1.6 m long. Find the height of the lamp-post. 3

23. Find the coordinates of the points which divide the segment joining A(−2, 2) and B(2, 8) into four equal parts. 3

OR

If $P(2, p)$ is the mid-point of the segment joining $A(6, -5)$ and $B(-2, 11)$, find p. Also find the length AB. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

24. If the sum of the first 6 terms of an AP is 36 and that of the first 16 terms is 256, find the sum of the first 10 terms. Also find its 20th term. 5

OR

Case study: Subba Rao started work in 1995 at an annual salary of ₹5000 and received an increment of ₹200 each year. 5

(i) In which year did his income reach ₹7000?

(ii) Find his total earnings in the first 11 years.

(iii) A student claims his salary in the 20th year was ₹9000. Justify or refute.

25. E is a point on side AD produced of a parallelogram ABCD and BE intersects CD at F. 5

(i) Show that $\triangle ABE \sim \triangle CFB$.

(ii) If $AB = 8$ cm, $BC = 9$ cm and $AE = 12$ cm, find CF.

OR

If AD and PM are medians of triangles ABC and PQR respectively, where $\triangle ABC \sim \triangle PQR$, prove that $\frac{AB}{PQ} = \frac{AD}{PM}$. 5

END OF QUESTION PAPER — **Total: 50 Marks**